CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 6: Triangles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Basic Proportionality Theorem

1. In $\triangle ABC$, D lies on AB and E lies on AC with $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, find x. CBSE 2026, 2 marks

SOLUTION

By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$.

$\frac{x}{x-2} = \frac{x+2}{x-1}$, so $x(x-1) = (x+2)(x-2)$

$x^2 - x = x^2 - 4$

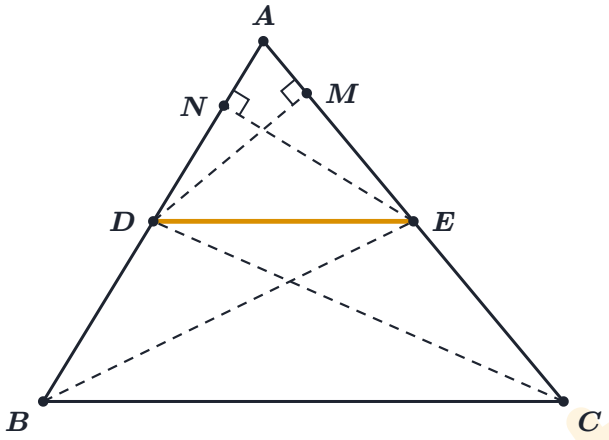
Answer: $x = 4$

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2. State the Basic Proportionality Theorem and prove it. CBSE 2026, 5 marks

SOLUTION

Statement: If a line is drawn parallel to one side of a triangle and cuts the other two sides at distinct points, it divides those two sides in the same ratio.



Given: $\triangle ABC$ with $DE \parallel BC$, where D is on AB and E is on AC. **To prove:** $\frac{AD}{DB} = \frac{AE}{EC}$.

Construction: Join BE and CD. Draw $EN \perp AB$ and $DM \perp AC$.

$$\frac{\text{ar}(ADE)}{\text{ar}(BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN} = \frac{AD}{DB} \quad \text{... (1)}$$

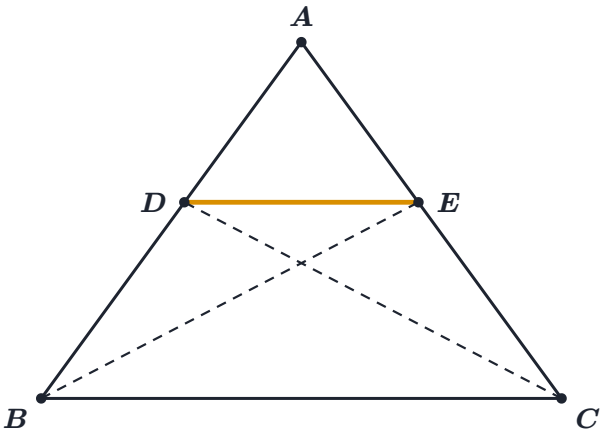
$$\frac{\text{ar}(ADE)}{\text{ar}(CED)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC} \quad \text{... (2)}$$

$\triangle BDE$ and $\triangle CED$ stand on the same base DE and lie between the same parallels DE and BC, so $\text{ar}(BDE) = \text{ar}(CED)$ ... (3)

From (1), (2) and (3): $\frac{AD}{DB} = \frac{AE}{EC}$. **Proved.**

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3. Prove that a line drawn parallel to one side of a triangle, cutting the other two sides at distinct points, divides those two sides in the same ratio. CBSE 2024, 5 marks



SOLUTION

Given: in $\triangle ABC$, $DE \parallel BC$, with D on AB and E on AC. To prove: $\frac{AD}{DB} = \frac{AE}{EC}$.

Construction: join BE and CD. Draw $EN \perp AB$ and $DM \perp AC$.

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN} = \frac{AD}{DB}$$

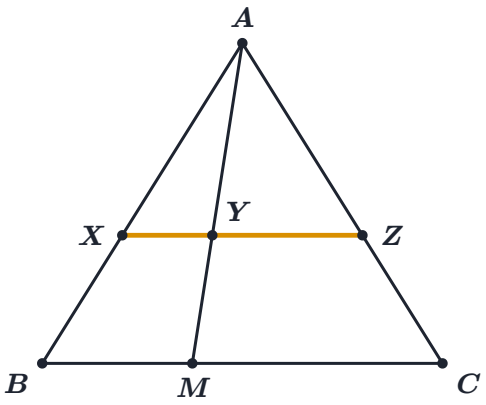
$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle CED)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC}$$

$\triangle BDE$ and $\triangle CED$ are on the same base DE and between the same parallels DE and BC, so $\text{ar}(\triangle BDE) = \text{ar}(\triangle CED)$.

Hence $\frac{AD}{DB} = \frac{AE}{EC}$.

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4. In $\triangle ABC$, X lies on AB and Z on AC with $XZ \parallel BC$. M is a point on BC, and AM meets XZ at Y. If $AZ = 3$ cm, $ZC = 2$ cm, $BM = 3$ cm and $MC = 5$ cm, find XY. CBSE 2023, 2 marks



SOLUTION

$XZ \parallel BC$, so $\dfrac{AX}{AB} = \dfrac{AZ}{AC} = \dfrac{3}{3+2} = \dfrac{3}{5}$.

In $\triangle ABM$, $XY \parallel BM$, so $\triangle AXY \sim \triangle ABM$ (AA).

$\dfrac{XY}{BM} = \dfrac{AX}{AB} = \dfrac{3}{5}$, so $XY = \dfrac{3}{5} \times 3$

Answer: $XY = 1.8$ **cm**

5. A line parallel to one side of a triangle meets the other two sides at distinct points. Prove that it divides those two sides in the same ratio. CBSE 2023, 5 marks

SOLUTION

This is the Basic Proportionality Theorem. In $\triangle ABC$, let $DE \parallel BC$ with D on AB and E on AC. Join BE and CD, and draw $EN \perp AB$ and $DM \perp AC$.

$\dfrac{\text{ar}(ADE)}{\text{ar}(BDE)} = \dfrac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN} = \dfrac{AD}{DB}$ and $\dfrac{\text{ar}(ADE)}{\text{ar}(CED)} = \dfrac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \dfrac{AE}{EC}$

$\triangle BDE$ and $\triangle CED$ are on the same base DE and between the same parallels, so their areas are equal.

Hence $\dfrac{AD}{DB} = \dfrac{AE}{EC}$**. Proved.** (The figure is with the 2026 question above.)

2 Criteria for similarity of triangles

6. $\triangle ABC \sim \triangle DEF$ with $2AB = DE$ and $BC = 8$ cm. Find EF. (a) 4 cm (b) 8 cm (c) 12 cm (d) 16 cm CBSE 2026, 1 mark

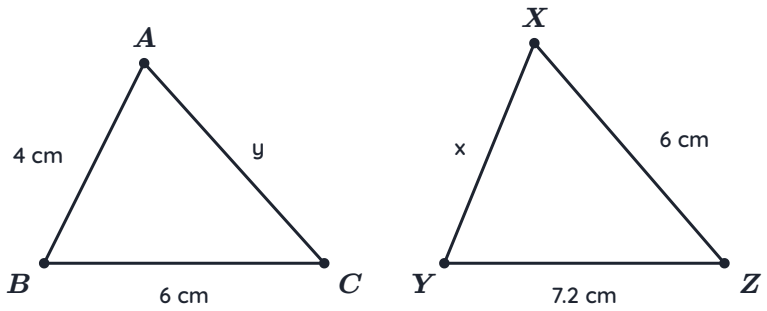
SOLUTION

Corresponding sides are proportional: $\dfrac{AB}{DE} = \dfrac{BC}{EF}$.

$\dfrac{AB}{DE} = \dfrac{1}{2}$, so $\dfrac{8}{EF} = \dfrac{1}{2}$ and $EF = 16$ cm.

Answer: (d)

7. $\triangle ABC \sim \triangle XYZ$ with $AB = 4$ cm, $BC = 6$ cm, $AC = y$ cm, $XY = x$ cm, $YZ = 7.2$ cm and $XZ = 6$ cm. Find x and y. CBSE 2026, 2 marks



SOLUTION

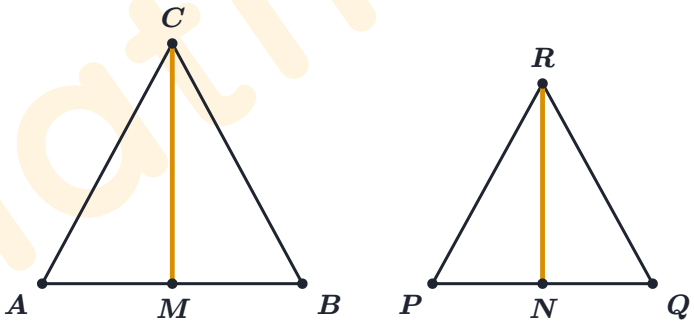
$\frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ}$, and $\frac{BC}{YZ} = \frac{6}{7.2} = \frac{5}{6}$

$\frac{4}{x} = \frac{5}{6}$, so $x = \frac{24}{5} = 4.8$

$\frac{y}{6} = \frac{5}{6}$, so $y = 5$

Answer: $x = 4.8$, $y = 5$

8. CM and RN are medians of $\triangle ABC$ and $\triangle PQR$, and $\triangle ABC \sim \triangle PQR$. Prove that (i) $\triangle AMC \sim \triangle PNR$ (ii) $\triangle CMB \sim \triangle RNQ$. CBSE 2026, 5 marks



SOLUTION

$\triangle ABC \sim \triangle PQR$, so $\angle A = \angle P$, $\angle B = \angle Q$ and $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}$.

M and N are mid-points, so $AB = 2AM = 2MB$ and $PQ = 2PN = 2NQ$. Hence $\frac{AM}{PN} = \frac{MB}{NQ} = \frac{AB}{PQ}$.

(i) In $\triangle AMC$ and $\triangle PNR$: $\frac{AM}{PN} = \frac{AC}{PR}$ and the included angles $\angle A = \angle P$. **So $\triangle AMC \sim \triangle PNR$ (SAS).**

(ii) In $\triangle CMB$ and $\triangle RNQ$: $\frac{MB}{NQ} = \frac{BC}{QR}$ and the included angles $\angle B = \angle Q$. **So $\triangle CMB \sim \triangle RNQ$ (SAS).**

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9. In $\triangle ABC$ and $\triangle DEF$, $\angle B = \angle E$, $\angle C = \angle F$ and $AB = 3DE$. Are the two triangles similar, congruent, both or neither? CBSE 2025, 1 mark

SOLUTION

Two pairs of angles are equal, so the triangles are similar by the AA criterion.

$AB = 3DE$, so the corresponding sides are not equal. The triangles are therefore not congruent.

Answer: similar but not congruent

10. $\triangle ABC \sim \triangle PQR$ with $AB = 6$ cm, $BC = 4$ cm, $AC = 8$ cm and $PR = 6$ cm. Find $PQ + QR$. CBSE 2025, 2 marks

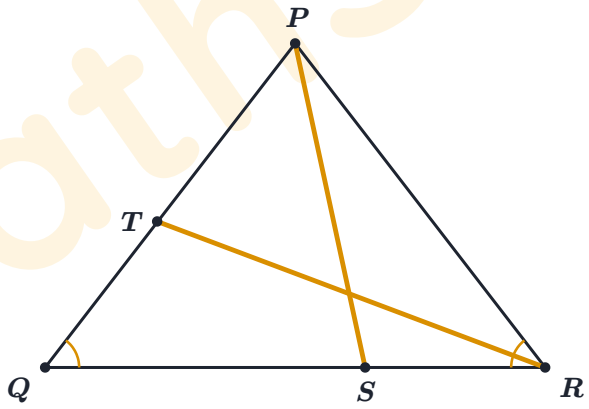
SOLUTION

Corresponding sides are in the same ratio: $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = \frac{8}{6} = \frac{4}{3}$

$PQ = 6 \times \frac{3}{4} = 4.5$ cm and $QR = 4 \times \frac{3}{4} = 3$ cm

Answer: $PQ + QR = 7.5$ cm

11. In $\triangle PQR$, S is a point on QR and T is a point on PQ. It is given that $\frac{QR}{QS} = \frac{QT}{PR}$ and $\angle PQR = \angle PRQ$. Prove that $\triangle PQS \sim \triangle TQR$. CBSE 2025, 2 marks



SOLUTION

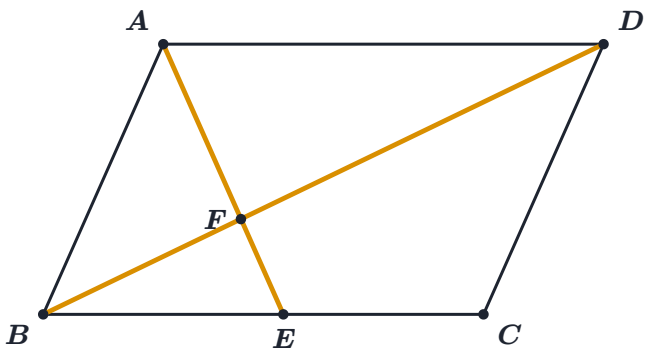
$\angle PQR = \angle PRQ$, so the sides opposite these angles are equal: $PR = PQ$.

Put PQ in place of PR in the given relation: $\frac{QR}{QS} = \frac{QT}{PQ}$, that is $\frac{QT}{QR} = \frac{PQ}{QS}$.

In $\triangle PQS$ and $\triangle TQR$: $\frac{PQ}{TQ} = \frac{QS}{QR}$ and the included angle $\angle Q$ is common.

So $\triangle PQS \sim \triangle TQR$ by the SAS criterion.

12. $ABCD$ is a parallelogram and E is a point on BC. The diagonal BD meets AE at F. Prove that $DF \times EF = FB \times FA$. CBSE 2025, 5 marks



SOLUTION

In $\triangle DFA$ and $\triangle BFE$: $\angle DFA = \angle BFE$ (vertically opposite angles).

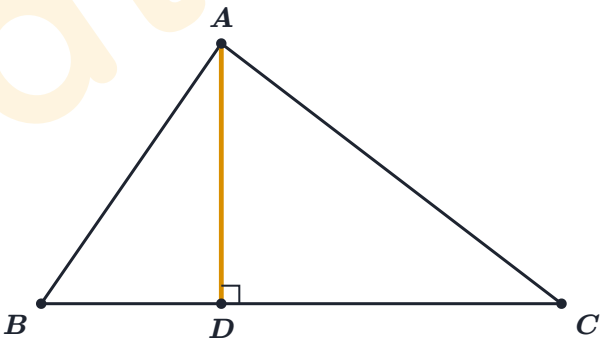
$AD \parallel BC$ and BD is a transversal, so $\angle ADF = \angle EBF$ (alternate angles).

So $\triangle DFA \sim \triangle BFE$ by the AA criterion.

Corresponding sides are proportional: $\frac{DF}{BF} = \frac{FA}{FE}$.

Cross-multiplying, $DF \times EF = FB \times FA$. **Proved.**

13. In $\triangle ABC$, D is a point on BC with $AD \perp BC$ and $AD^2 = BD \times DC$. Prove that $\angle BAC = 90°$. CBSE 2025, 5 marks



SOLUTION

In right triangle ADB: $AB^2 = AD^2 + BD^2$ ... (1)

In right triangle ADC: $AC^2 = AD^2 + DC^2$ ... (2)

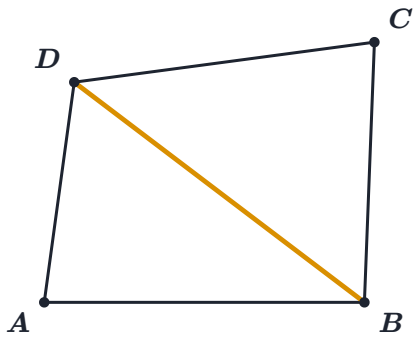
Adding (1) and (2): $AB^2 + AC^2 = 2AD^2 + BD^2 + DC^2$

Since $AD^2 = BD \times DC$: $AB^2 + AC^2 = BD^2 + 2\,BD \times DC + DC^2 = (BD + DC)^2 = BC^2$

By the converse of the Pythagoras theorem, the angle opposite BC is a right angle.

So $\angle BAC = 90°$. Proved.

14. In quadrilateral $ABCD$, the diagonal BD bisects both $\angle B$ and $\angle D$. Prove that (i) $\triangle ABD \sim \triangle CBD$ (ii) $AB = BC$. CBSE 2024, 2 marks



SOLUTION

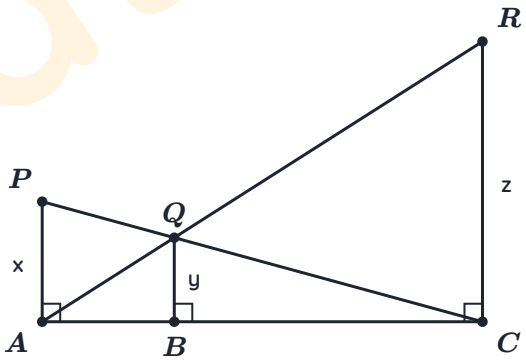
(i) In $\triangle ABD$ and $\triangle CBD$: $\angle ABD = \angle CBD$ (BD bisects $\angle B$) and $\angle ADB = \angle CDB$ (BD bisects $\angle D$).

So $\triangle ABD \sim \triangle CBD$ by the AA criterion.

(ii) Corresponding sides are proportional: $\frac{AB}{CB} = \frac{BD}{BD} = 1$

Hence $AB = BC$.

15. PA, QB and RC are all perpendicular to the line AC, where Q is the point in which PC and AR meet. If $AP = x$, $BQ = y$ and $CR = z$, prove that $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$. CBSE 2024, 5 marks



SOLUTION

In $\triangle CQB$ and $\triangle CPA$: $\angle C$ is common and $\angle CBQ = \angle CAP = 90°$, so $\triangle CQB \sim \triangle CPA$ (AA).

Hence $\frac{BQ}{AP} = \frac{BC}{AC}$, that is $\frac{y}{x} = \frac{BC}{AC}$.

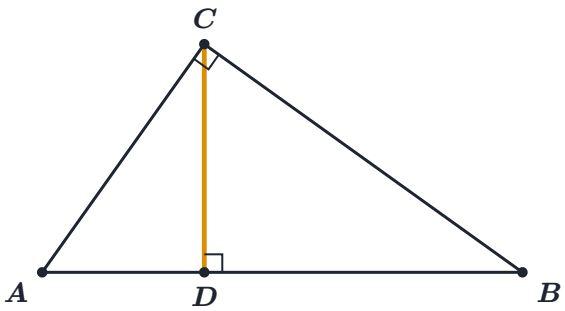
In $\triangle AQB$ and $\triangle ARC$: $\angle A$ is common and $\angle ABQ = \angle ACR = 90°$, so $\triangle AQB \sim \triangle ARC$ (AA).

Hence $\frac{BQ}{CR} = \frac{AB}{AC}$, that is $\frac{y}{z} = \frac{AB}{AC}$.

Adding: $\frac{y}{x} + \frac{y}{z} = \frac{BC + AB}{AC} = \frac{AC}{AC} = 1$

Dividing by y: $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$.

16. In $\triangle ABC$, $\angle ACB = 90^\circ$ and $CD \perp AB$. Prove that $CD^2 = BD \times AD$. CBSE 2019, 3 marks



SOLUTION

In $\triangle ADC$ and $\triangle CDB$: $\angle ADC = \angle CDB = 90^\circ$.

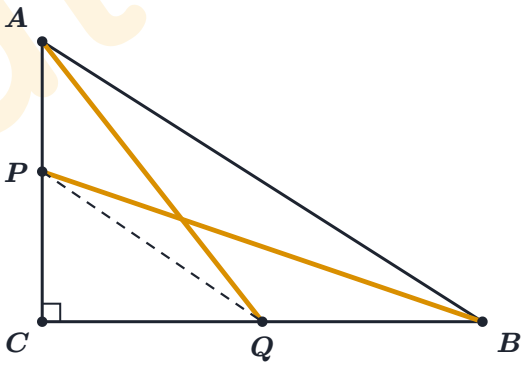
$\angle DAC = 90^\circ - \angle ACD$ (angle sum in $\triangle ADC$) and $\angle DCB = 90^\circ - \angle ACD$ (because $\angle ACB = 90^\circ$). So $\angle DAC = \angle DCB$.

Hence $\triangle ADC \sim \triangle CDB$ by the AA criterion.

$$\frac{AD}{CD} = \frac{CD}{BD}$$

So $CD^2 = BD \times AD$. Proved.

17. $\triangle ABC$ is right-angled at C. P is a point on CA and Q is a point on CB. Prove that $AQ^2 + BP^2 = AB^2 + PQ^2$. CBSE 2019, 3 marks



SOLUTION

Each of the triangles ACQ, PCB, ACB and PCQ has a right angle at C. Apply the Pythagoras theorem to each.

$AQ^2 = AC^2 + CQ^2$ and $BP^2 = PC^2 + BC^2$

Adding: $AQ^2 + BP^2 = (AC^2 + BC^2) + (PC^2 + CQ^2)$

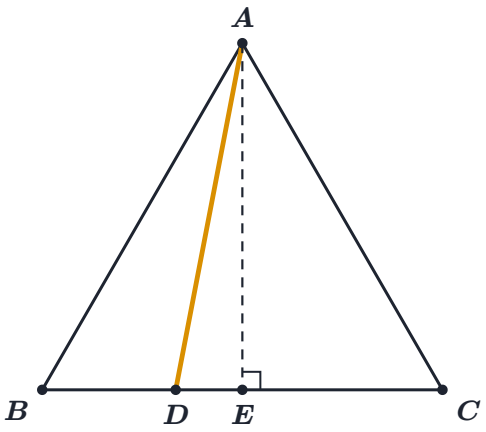
$AC^2 + BC^2 = AB^2$ and $PC^2 + CQ^2 = PQ^2$

So $AQ^2 + BP^2 = AB^2 + PQ^2$. Proved.

3 Right-angled triangles

18. ABC is an equilateral triangle and D is a point on BC with $BD = \frac{1}{3}BC$. Prove that $9AD^2 = 7AB^2$.

CBSE 2018, 4 marks



SOLUTION

Let each side be a. Draw $AE \perp BC$. In an equilateral triangle the altitude bisects the base, so $BE = \frac{a}{2}$.

$BD = \frac{a}{3}$, so $DE = BE - BD = \frac{a}{2} - \frac{a}{3} = \frac{a}{6}$.

In right triangle AEB: $AE^2 = a^2 - \frac{a^2}{4} = \frac{3a^2}{4}$

In right triangle AED: $AD^2 = AE^2 + DE^2 = \frac{3a^2}{4} + \frac{a^2}{36} = \frac{28a^2}{36} = \frac{7a^2}{9}$

So $9AD^2 = 7a^2 = 7AB^2$. Proved.

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